New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

FULL NAME: | SIGNATURE:

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Mock Midterm Exam

COURSE NAME:	Computer Programming I	COURSE NUMBER:	CS 102
EXAMINATION DURATION:	90 MINUTES	EXAM VERSION	MOCK 2
ADDITIONAL MATERIALS ALLOWED TO USE:	No additional materials or items are allowed to be used.		

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST! VIOLATION OF THE RULES CAN LEAD TO A LOSS OF POINTS.

- There are **7 problems** worth **100 points**. Write every answer **in the space provided**; if you need more space, use the back of the page and say so.
- Assume every code fragment is inside `main` with `#include <iostream>`, `<iomanip>`, `<cmath>` and `using namespace std;`. When a fragment shows its own functions, the lines after `// in main:` are inside `main`.
- Points are shown in brackets — roughly one minute per point. Partial credit is given for clear reasoning. Small syntax slips in your code cost little; wrong logic costs a lot.
- Put your signature and write your ID on every page.
- Follow the University Examination Rules.

FOR INSTRUCTOR USE ONLY (DO NOT WRITE ANYTHING):

Problem	Type of Questions	Points	Score
1	Short Answer	**14**	
2	Number Systems and Data	**12**	
3	Algorithms	**12**	
4	What Does It Print?	**20**	
5	Debugging	**12**	
6	Complete the Program	**10**	
7	Write the Code	**20**	
Total		**100**	

SIGNATURE: _____ ID: _____

Problem 1: Short Answer (14 points)

For (a)–(e) circle TRUE or FALSE. For (f)–(g) write a short answer on the line.

(a) [2 points] `cout << 'A' + 2;` prints `C`. TRUE FALSE

(b) [2 points] `int x = 7; if (x = 3) cout << "yes"; else cout << "no";` compiles and prints `yes`. TRUE FALSE

(c) [2 points] A function `void f(int x, int y = 2)` can be called as `f()`. TRUE FALSE

(d) [2 points] In C++, `7 % 2.0` is `1.0`. TRUE FALSE

(e) [2 points] `double d = 5 / 2;` stores the value 2.5 in `d`. TRUE FALSE

(f) [2 points] After this code, what is the value of `count`? Show how you counted.

```
int count = 0;
for (int i = 1; i <= 30; i *= 3)
    for (int j = i; j <= 30; j += i)
        count++;
```

(g) [2 points] Why does `for (double x = 1; x != 0; x -= 0.1)` never stop? How should the condition be written?

Problem 2: Number Systems and Data (12 points)

Show your work for (a)–(d).

(a) [2 points] Compute `0x2B3` + `0b1101` in decimal and in binary.

(b) [2 points] Convert $5D2_{16}$ to octal (hint: go through binary).

(c) [2 points] Convert 34021 (base 5) to decimal and then to octal.

(d) [2 points] How many bits does 500 000 need? What is the largest number with that many bits?

(e) [2 points] What does this print? (`'a'` is 97 in ASCII.)

```
char c = 'x';
for (int i = 0; i < 3; i++) c = (c - 'a' + 9) % 26 + 'a';
cout << c << (int)c;
```

(f) [2 points] Assume `int` is 32 bits and overflow wraps around. What are `b` and `c`, and why are they different?

```
int a = 70000;
long long b = a * a;
long long c = (long long)a * a;
```

Problem 3: Algorithms: Pseudocode, Tracing, Flowcharts (12 points)

(a) [6 points] Trace this pseudocode: one row per pass (values at the end of the pass). Write what it prints, and say in one sentence what `r` and `c` contain for any input without a zero digit.

```
SET n = 84693
SET r = 0
SET c = 0
WHILE n > 0 DO
    SET d = n MOD 10
    IF d MOD 3 = 0 THEN
        SET r = r * 10 + d
    ELSE
        SET c = c + d
    SET n = n DIV 10
PRINT r, c
```

Pass	d	n	r	c
before	—	84693	0	0
1				
2				
3				
4				
5				

Prints: _____

r is: _____

c is: _____

(b) [3 points] This pseudocode should print the sum of the even digits of n (for 48613 it must print 18). It contains two faults: find and correct both.

```
INPUT n
SET sum = 0
WHILE n > 0 DO
    SET d = n DIV 10
    IF d MOD 2 = 0 THEN
        SET sum = d
    SET n = n DIV 10
PRINT sum
```

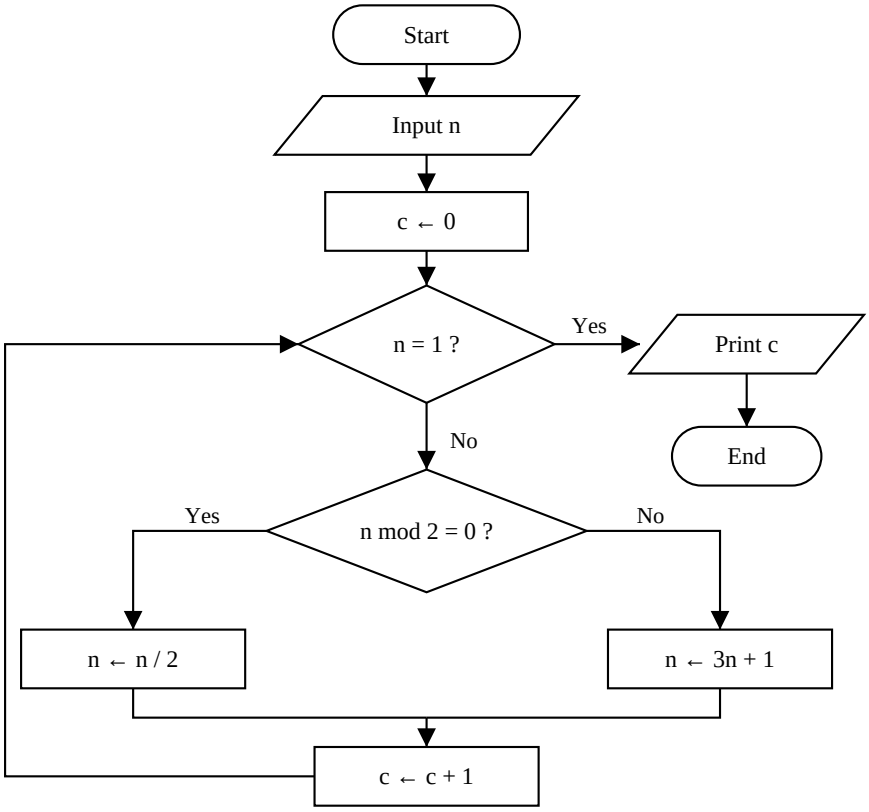
Fault 1: _____

Fix 1: _____

Fault 2: _____

Fix 2: _____

(c) [3 points] What does this flowchart print for n = 6? For n = 5? What happens for n = 0, and why?



n = 6: _____

n = 5: _____

n = 0: _____

Problem 4: What Does It Print? (20 points)

Write exactly what each fragment prints (mind the spaces). If a fragment does not compile, write ERROR and explain why.

(a) [4 points]

```
int a = 7, b = 2;
double x = a / b + (double)a / b * 0.5 + a % b;
int y = ++a * 2 - b--;
bool z = (a < b) && (b = 5) || !(a % 2);
cout << x << " " << y << " " << a << " " << b << " " << z << endl;
```

Output: _____

If ERROR, why: _____

(b) [4 points]

```
char op = 'c';
int v = 4;
switch (op - 1) {
    case 'a': v += 1;
    case 'b': v *= 5;
    case 'c': v = v > 10 ? v - 15 : v + 15;
    case 'd': v += 2; break;
    default: v = 0;
}
cout << v << endl;
```

Output: _____

If ERROR, why: _____

(c) [4 points]

```
int s = 0;
for (int i = 1; i <= 5; i++) {
    int j = i;
    while (j > 0) {
        j -= 2;
        if (j == 1) continue;
        if (i % 3 == 0) break;
        s += i * j;
    }
}
cout << s << endl;
```

Output: _____

If ERROR, why: _____

(d) [4 points]

```
int total = 0;
for (int i = 1; i <= 3; i++) {
    int sq = i * i;
    total += sq;
}
cout << total << " " << sq << endl;
```

Output: _____

If ERROR, why: _____

(e) [4 points]

```
int f(int& a, int b = 2) {
    a += b;
    return a * b;
}
int f(double a) {
    return a * 3;
}
// in main:
int p = 5, q = 1;
int r1 = f(p);
int r2 = f(q, p);
cout << r1 << " " << r2 << " " << p << " " << q << " " << f(p / 2.0) << endl;
```

Output: _____

If ERROR, why: _____

Problem 5: Debugging (12 points)

This program should read two numbers a and b (in any order) and print how many numbers between them (both included) have only odd digits (like 7, 15, 391 — but not 12 or 30), and the sum of those numbers. For the input `39 15` it must print `8 226` (15, 17, 19, 31, 33, 35, 37, 39). It contains exactly four bugs; the program compiles, so all four are logic errors. For each bug give the line number(s), what is wrong, and the corrected code.

(a) [12 points]

```
 1 #include <iostream>
 2 using namespace std;
 3
 4 bool allOdd(int n);
 5
 6 int main() {
 7     int a, b, count = 0, sum = 0;
 8     cin >> a >> b;
 9     if (a < b) {
10         int t = a;
11         a = b;
12         b = t;
13     }
14     for (int x = a; x < b; x++) {
15         if (allOdd(x)) {
16             count++;
17         }
18         sum += x;
19     }
20     cout << count << " " << sum << endl;
21     return 0;
22 }
23
24 bool allOdd(int n) {
25     while (n > 0) {
26         int d = n / 10;
27         if (d % 2 == 0) {
28             return false;
29         }
30         n /= 10;
31     }
32     return true;
33 }
```

Line	What is wrong	Corrected code

Problem 6: Complete the Program (10 points)

The program reads an integer whose decimal digits are all 0 or 1 (at most 9 digits), a binary number written like 101101, and prints its value in decimal and how many 1s it contains. Input `101101` must print `45 (4 ones)`; input `0` must print `0 (0 ones)`. Fill in the five gaps.

(a) [10 points]

```
void fromBinary(int bits, (1) value, int& ones);

int main() {
    int bits;
    cin >> bits;
    int v, k;
    (2) ;
    cout << v << " (" << k << " ones)" << endl;
    return 0;
}

void fromBinary(int bits, int& value, int& ones) {
    value = 0;
    ones = 0;
    int power = 1;
    while (bits > 0) {
        int d = bits % 10;
        if (d == 1) {
            ones++;
        }
        value += (3) ;
        power (4) ;
        bits (5) ;
    }
}
```

(1)	(2)	(3)	(4)	(5)

Problem 7: Write the Code (20 points)

(a) [10 points] Write three functions. `int countDigits(int n)` returns the number of digits of n ≥ 1. `int digitAt(int n, int k)` returns the k-th digit of n counted from the left, starting at 1 (`digitAt(48215, 2)` is 8, `digitAt(7, 1)` is 7). `bool isPalindrome(int n)` returns true when the digits of n read the same from both ends (12321 and 4004: true; 1231: false). `digitAt` and `isPalindrome` must use the functions above them. Write only the three function definitions; for full marks, do not use `string`, `to_string`, `pow` or any other built-in/library function.

(b) [10 points] Write a complete program (`main` only). It reads n (you may assume 1 ≤ n ≤ 9) and prints an n × n table in which every cell holds its distance from the border: the outer ring holds 1, the next ring holds 2, and so on. Every number takes 4 columns (`setw(4)`). After the table print the sum of all numbers on a line starting with `Total:`, then the sum of the main diagonal (top left to bottom right) on a line starting with `Diagonal:`. Use nested `for` loops and compute every number from its row and column — no arrays.

Input 4:

```
   1   1   1   1
   1   2   2   1
   1   2   2   1
   1   1   1   1
Total: 20
Diagonal: 6
```

END OF EXAM